**SDC HACKATHON 2026**

# Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | XH-Cat |
| **Project / Solution Name** | FutureReady |
| **Name (Leader)** | YIP GUI RONG |
| **Email & Phone (Leader)** | guirong05@gmail.com |
| **University Affiliation(s)** | Multimedia University |
| **Live Prototype URL** | https://futures.grong.my |
| **Code Repository URL (public / access granted to judges)** | https://github.com/guirong0109/XH-Cat_MyFuture |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

Browser (plain JavaScript ES modules, three.js landing intro; English, Bahasa Melayu, Chinese) -> Node.js 24 + Express 5 JSON API on the team's computer -> Cloudflare Tunnel and DNS -> https://futures.grong.my.
CV upload: pdf.js reads the PDF in the browser; the server runs a rules reader and the DeepSeek deepseek-flash chat model (JSON mode). An AI item is kept only if it quotes the CV word for word; the user confirms each one.
Career matching, the Future Skills Radar and the study pathway are arithmetic on bundled official data (O*NET 31.0, eMASCO, ISCED-F, BLS, ILO, Anthropic Economic Index, OpenAlex, DOSM, UN SDG), each with a SHA-256 manifest. No language model is used for matching.
AI news: every 3 hours DeepSeek Search looks for news in 10 study fields and deepseek-flash drops results that are not news about AI in that field; caps of 10 per run and 80 per day.
Storage: accounts, sessions, profiles and organisations in SQLite (node:sqlite); news in a JSON file.

### 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| JavaScript (ES2023), HTML, CSS; Python 3.14 (data pipeline, standard library only) | Languages of the app and of the data build scripts | Open standards; Python PSF License |
| Node.js 24.16 | Server runtime (also its built-in node:sqlite) | MIT |
| Express 5.3.0 | Web server and JSON API | MIT |
| pdfjs-dist 6.4.299 (pdf.js) | Reads uploaded CV PDFs in the browser | Apache-2.0 |

| Name & Version | Purpose in Project | License |
|---|---|---|
| three.js 0.186.1 | 3D landing intro | MIT |
| Vitest 5.0.5 | Automated tests (1,765 tests) | MIT |
| Geist, Geist Mono, Instrument Serif (Google Fonts) | Typefaces | SIL Open Font License 1.1 |
| cloudflared 2025.8.1 | Runs the Cloudflare Tunnel to the live site | Apache-2.0 |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| DeepSeek chat API (deepseek-flash), DeepSeek | Reads each uploaded CV (JSON mode); relevance check on AI news results | Paid (team's own account, about US$0.002 to US$0.004 per CV) |
| DeepSeek Search API, DeepSeek | Finds AI news for 10 study fields every 3 hours | Paid (US$0.75 per 1,000 searches; capped at 80 a day) |
| OpenAlex API, OurResearch | Research counts for the Radar (fetched by the data pipeline, bundled) | Free |
| UN SDG API, UN Statistics Division | SDG goals and targets text (fetched by the pipeline, bundled) | Free |
| OpenDOSM data, Department of Statistics Malaysia | Job Market Insights and labour figures (fetched by the pipeline, bundled) | Free |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Cloudflare DNS and proxy (grong.my) | Domain, HTTPS certificate and caching for futures.grong.my | Free plan / global edge |
| Cloudflare Tunnel (cloudflared) | Publishes the app from the team's computer with no open router ports | Free plan / global edge |
| Team's own computer (self-hosted) | Runs Node.js, SQLite and the news store for the live site | Self-hosted / Malaysia |
| GitHub | Code repository and commit history | Free plan |

| Service & Provider | Purpose in Project | Plan / Region |
| --- | --- | --- |
| Cloudflare D1 and R2 | Supported by the code as an alternative store, not used for the live site | Not used |

## 4. AI Tools, Models & Datasets

### 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| deepseek-flash (DeepSeek chat API) | DeepSeek | Pre-trained, called through the API (no fine-tuning) | DeepSeek API terms of service |
| DeepSeek Search | DeepSeek | Pre-trained search service, called through the API | DeepSeek API terms of service |
| BAAI/bge-m3 (embedding model) | BAAI, Hugging Face | Pre-trained; evaluated in the preliminary round only, not in the final build | MIT |

### 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| O*NET 31.0 database | onetcenter.org | Public | CC BY 4.0 | N |
| eMASCO / MASCO 2020 occupations | emasco.mohr.gov.my | Public | Public government reference | N |
| ISCED-F 2013, Statistics Canada concordance, O*NET CIP crosswalk | UNESCO UIS; statcan.gc.ca; onetcenter.org | Public | Statistics Canada Open Licence; CC BY 4.0 | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| BLS SOC 2018/2010 and SOC/ISCO-08 crosswalks | bls.gov | Public | US government work (public domain) | N |
| ILO Working Paper 140 GenAI exposure scores | ilo.org | Public | CC BY 4.0 | N |
| Anthropic Economic Index (2026-06-26) | huggingface.co/datasets/Anthropic/EconomicIndex | Public | CC-BY | N |
| OpenAlex research counts | openalex.org | Public | CC0 | N |
| DOSM Job Market Insights | open.dosm.gov.my | Public | Public DOSM release | N |
| UN SDG goals and targets | unstats.un.org/sdgapi | Public | UN material | N |
| Course and accredited programme picks (data/pathway/picks.json) | Team-collected links to public course and MQA programme pages | Self-collected | Team-made list; each page keeps its own terms | N |
| Test CVs and organisation rosters | Written by the team for tests and the CV evaluation | Synthetic | Team-made | N (made-up people) |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

| **Tool & Version** | **Used For** | **Extent of Use (minor / moderate / major)** |
|---|---|---|
| Claude Code (Anthropic, Claude models) | Writing code, tests, specs, plans and documentation, and code reviews, under the team's direction | Major |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

Every feature started from a design and architecture diagram the team leader approved, then a test-first plan. AI-written code had to pass the automated tests (1,765 Vitest and 49 Python tests) before each commit, was checked by whole-branch code reviews, and was tried in the browser on the local copy before going live. Every data source is recorded with its licence and a hash in data/SOURCES.md. The team leader decided every design and merge and takes responsibility for the result.

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☒Yes ☐ No

If yes, list them below:

| **Sponsor** | **Resource (API / credits / platform / data)** | **How It Was Used in the Project** |
|---|---|---|
| Replit | Platform | Web Deploy and Hosting in the preliminary stage |
| Runpod | Credits | To identify the local LLM selected |
| None in the final build | Not applicable | Not applicable |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Not applicable to the final build: it uses no SDC sponsor resource. An SDC sponsor resource was used only in the preliminary round, in a separate repository; that work is not part of this prototype. The final build runs on the team's own DeepSeek account and free Cloudflare services.

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

Collected: the profile a user types or confirms (skills, education, interests), an optional uploaded CV, and, only if they sign up, an email, a password and saved profiles. A guest's progress stays in their browser.
Sent to third parties: the text of an uploaded CV goes to the DeepSeek API to be read; nothing else is sent to an AI service. The news search sends only field names, never user data.
Protection: passwords are hashed with scrypt; sessions use HttpOnly cookies; the site is served over HTTPS through Cloudflare; API keys stay in a local .env file outside the repository. The profile sent for matching is used for one answer and is never stored or logged. An organisation's roster is stored under its owner's account only. Deleting an account deletes its profiles and organisation data.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

AI reading: the CV model can miss or invent items; we keep only items quoted word for word from the CV and the user confirms each one. AI news relevance can still be wrong, so every item links to its source.
Single provider: both AI features depend on DeepSeek; if it fails, the rules-only CV reader still works and the last news list stays on screen. Daily caps limit cost but can delay news.
Data: O*NET is US-based, mapped to Malaysian eMASCO roles through official code crosswalks; a broken chain shows "not available" rather than a guess. Snapshots can age; the screen shows each source date.
Hosting: the live site runs on one computer, so it is offline if that computer is off. The student pilot has not been run yet.

## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

YIP

1 November 2026